

Quantitative Data Appendices: Tang Dynasty and Abbasid Caliphate

These comprehensive appendices compile quantitative historical data for the **Tang Dynasty (618-907 CE)** and **Abbasid Caliphate (750-1258 CE)** as part of the Infonomics Project CAS Framework. Data is organized by Rise/Peak/Decline phases with explicit confidence assessments and source documentation.

APPENDIX A: Tang Dynasty Quantitative Data (618-907 CE)

Executive Summary

The Tang Dynasty represents one of medieval China's most prosperous eras, with peak population estimates of **50-75 million**, the world's largest city (**Chang'an at 1-2 million**), and sophisticated institutional systems including the **Tang Code (502 articles)** and the **keju examination system**. The An Lushan Rebellion (755-763) marks the critical phase transition, causing **13-36 million deaths** (highly disputed) ([Wikipedia](#)) and transforming military and fiscal structures fundamentally.

1. Economic Indicators

1.1 GDP Estimates (Maddison 1990 International Dollars)

Period	GDP per Capita	Confidence	Source/Notes
Tang Dynasty (overall)	~\$480	LOW	Maddison (2007); backward extrapolation, not measurement
Han Dynasty benchmark (1 CE)	\$450	LOW	Maddison (2007) baseline comparison
Song Dynasty (960-1279)	Higher than Tang	MODERATE	Proto-industrialization consensus

CRITICAL METHODOLOGICAL NOTE: Maddison's pre-1700 China estimates are **heavily disputed**. Kent Deng & Patrick O'Brien (LSE Working Paper 229, 2016) argue these estimates "are not fit for purpose" for measuring imperial China's GDP. Paul Bairoch criticized Maddison for underestimating Asian per-capita income before the 19th century. **No direct Tang-specific GDP calculation exists** - all figures are backward extrapolations.

World GDP Share Claims: Various popular sources claim Tang China held 58% of world GDP - **this figure is HIGHLY QUESTIONABLE** and lacks rigorous scholarly support.

1.2 Trade Volume and Routes

Route	Status	Period	Confidence
Silk Road (Central Route)	Open and flourishing	639-907 (interrupted 678-699)	HIGH
Southern Maritime Route	Active	Throughout	MODERATE
Postal service network	~32,100 km	Peak Tang	HIGH

DATA GAP: No reliable quantitative monetary values exist for Tang Silk Road trade volumes. Trade is documented qualitatively through foreign merchant communities (Persian, Arab, Sogdian, Indian) in major cities.

Key Exports: Silk, porcelain, paper, lacquerwares, tea
Key Imports: Horses, jade, amber, textiles (cotton), ivory, spices, precious metals

1.3 Urbanization Rates

Period	Urbanization Rate	Confidence	Source
Early Tang	5%	MODERATE	Rozman (1973)
Tang Dynasty (overall)	4-5%	MODERATE	Rozman (1973)
Late Tang (Adshead)	Up to 20%	LOW	Likely overstated

Rural/agrarian areas comprised **80-90%** of Tang population throughout the dynasty.

1.4 Major Economic Centers

City	Population (Peak c.750)	Status	Confidence
Chang'an	800,000-1,000,000 (walls); 2-3 million (metro)	World's largest city	HIGH
Luoyang	~1,000,000	Eastern capital	MODERATE
Yangzhou	Major trading hub	Salt monopoly HQ	LOW
Guangzhou (Canton)	Major port	Principal maritime port	LOW

Urban Infrastructure (754 Census): 1,859 cities, 321 prefectures, 1,538 counties. Fourteen urban regions had more than 170,000 households.

2. Demographic Indicators

2.1 Total Population by Phase

Year	Households	Population	Phase	Confidence	Source
609 (Sui)	9 million	~50 million	Pre-Tang	HIGH	<i>Book of Sui</i> census
618 (Founding)	—	~50 million	Rise	MODERATE	Population disrupted
700	—	~50-55 million	Rise	MODERATE	Census records
742	8.9 million	50M registered; 70-75M actual	Peak	MODERATE	<i>New Book of Tang</i> ; Adshead adjustment
754	8,914,709	52,919,309	Peak	HIGH	Last pre-rebellion census
764	2,900,000	16,900,000 (registered)	Decline	MODERATE	Administrative collapse
907	—	~80 million	End	LOW	Estimate

KEY DISPUTE: Patricia Ebrey notes population remained stable at ~50-60M from Han through Tang. S.A.M. Adshead argues higher figure of ~75M by 750.

2.2 An Lushan Rebellion Casualties (755-763 CE)

Estimate	Source	Notes	Confidence
36 million	Census comparison	Difference: 52.9M → 16.9M registered	LOW
13 million	Matthew White (Atrocityology, 2011 revised)	More defensible estimate	MODERATE
20,000-30,000	Siege of Suiyang (757)	Including cannibalized	HIGH

SCHOLARLY CONSENSUS: The 36 million figure is problematic. Post-rebellion census reflects administrative collapse, not actual deaths. Many unregistered due to tax collection breakdown. **Better estimate:**

13 million deaths (6.5% of world population) - still among deadliest pre-modern conflicts.

2.3 Migration Patterns

Period	North Population	South Population	Confidence
Early Tang (618)	75%	25%	MODERATE
Late Tang (907)	50%	50%	MODERATE

Major southward migration occurred post-An Lushan Rebellion.

3. Financial Indicators

3.1 Tang Copper Cash (Kaiyuan Tongbao)

Attribute	Value	Confidence
Introduction	621 CE (4th year of Wude)	HIGH
Standard Weight	2.4 zhu; 10 coins = 1 liang	HIGH
Actual Weight	~3.21-3.75 grams	HIGH
Diameter	~2.4 cm / 22 mm	HIGH
Official Alloy	83% Cu, 15% Pb, 2% Sn	MODERATE
Actual Composition (2004 analysis)	70.21% Cu, 17.85% Pb, 8.64% Sn	HIGH

3.2 Currency Debasement Patterns

Period	Pattern	Confidence
621-750s CE	Coins adhered to ~83% copper specification	MODERATE
Post-755 CE	Lead content increased up to 33%; lighter coins	HIGH
845 CE (Huichang)	Buddhist temple bronzes melted; "poor workmanship"	HIGH

Counterfeiting: Prevalent despite death penalty; lead content >36% indicates counterfeits.

Flying Cash (feiqian): Developed 804-820 CE; exchange fee 100 wen per 1,000 wen; officially accepted 812 CE. Wikipedia

3.3 Government Fiscal Capacity - Revenue by Phase

Period	Revenue Source	Amount (strings of cash)	% of Total	Confidence
Pre-758	Salt monopoly	~400,000/year	Small	MODERATE
780	Salt monopoly	6,000,000/year	>50% of cash revenue	HIGH
Late Tang	Salt monopoly	>50% of annual revenue	—	HIGH

3.4 Tax System Evolution

Zuyongdiao System (Pre-780):

Tax	Type	Amount
Zu 租	Grain tax	2 dan (119 liters)
Diao 調	Textile tax	2 zhang (6.22m) silk + ~120g thread
Yong 庸	Corvée labor	20 days/year

Two-Tax System (Liangshuifa) - 780 CE:

- **Architect:** Chancellor Yang Yan (727-781) under Emperor Dezong
- **Structure:** Household tax (hushui) + Land tax (dishui)
- **Collection:** Twice yearly (summer before June; autumn before November)
- **Payment:** Primarily cash (not goods/labor)
- **Assessment:** 9 tax brackets based on household wealth
- **Duration:** Remained until Ming dynasty single-whip reform (16th century)
- **Confidence:** HIGH

3.5 Defense Budget

Year	Budget (strings)	Notes	Confidence
712	2 million	Low fubing maintenance	HIGH
742	12 million	6x increase with professional army	HIGH
755	15 million	Peak pre-rebellion	HIGH

7-fold increase from 712 to 755 due to shift from militia to professional army (Source: *Tongdian*)

4. Institutional Indicators

4.1 Complete Emperor List (618-907)

#	Temple Name	Personal Name	Reign	Years	Phase	Succession
1	Gaozu	Li Yuan	618-626	8	Rise	Founder
2	Taizong	Li Shimin	626-649	23	Rise	Coup (Xuanwu Gate)
3	Gaozong	Li Zhi	649-683	34	Rise	Peaceful
4	Zhongzong	Li Xian	684	<1	Rise	Deposed
5	Ruizong	Li Dan	684-690	6	Rise	Deposed
—	Wu Zetian	Wu Zhao	690-705	15	Zhou	Usurpation
6	Zhongzong (rest.)	Li Xian	705-710	5	Rise	Poisoned
7	Shang	Li Chongmao	710	<1	Rise	Deposed
8	Ruizong (rest.)	Li Dan	710-712	2	Rise	Abdicated
9	Xuanzong	Li Longji	712-756	44	Peak	Forced abdication
10	Suzong	Li Heng	756-762	6	Decline	Peaceful
11-23	[13 more emperors]	—	762-907	—	Decline	Mixed

Total Emperors: 20-21 (excluding/including Wu Zetian)
Confidence: HIGH (Source: Twitchett, *Cambridge History of China* Vol. 3)

4.2 Succession Statistics by Phase

Phase	Period	Emperors	Total Years	Avg. Reign	Violent %
Rise	618-712	8	94	11.8 years	~50%
Peak	712-755	1	44	44 years	0%
Decline	755-907	14	152	10.9 years	~50%
Overall	618-907	23	289	12.6 years	—

4.3 Tang Code (唐律疏義)

Attribute	Value	Confidence
First Code (Wude lü)	624 CE	HIGH
Zhenguan Revision	637 CE	HIGH
Definitive Version	652-653 CE (Yonghui lü shuyi)	HIGH
Total Articles	502 articles in 12 sections (30 juan)	HIGH
Compilers	Zhangsun Wuji, Fang Xuanling	HIGH

Source: Wallace Johnson, *The T'ang Code* (Princeton UP, 1979)

Five Punishments (Wu Xing): Light beating (10-50 blows), Heavy beating (60-100), Penal servitude (1-3 years), Exile (2,000-3,000 li), Death (strangulation/decapitation)

4.4 Examination System (Keju)

Metric	Value	Confidence	Source
Total Jinshi (618-907)	6,585	MODERATE-HIGH	Chen/Kung/Ma (Chicago)
Average per year	~23	MODERATE	Calculated
Candidates annually	1,000-2,000	MODERATE	Wikipedia

Metric	Value	Confidence	Source
Pass rate	1-2%	HIGH	Multiple sources

Official Selection Routes:

Method	Est. %	Notes
Examination (jinshi)	15-20%	Minority but growing
Yin privilege (shadow)	50-60%	Relatives of officials
Recommendation	20-30%	Traditional aristocratic

4.5 Administrative Structure

Provincial Administration (740):

Division	Number	Confidence
Circuits (Dao)	15	HIGH
Prefectures (Zhou)	328	HIGH
Counties (Xian)	1,573	HIGH

Bureaucratic Ranks: 9 principal ranks × 2 (zheng/cong) = 18 half-ranks

5. Military Indicators

5.1 Army Systems by Phase

Fubing Militia System (618-749):

Metric	Value	Confidence
Total Units	~600-634 zhechongfu	HIGH
Soldiers per Unit	800-1,200	HIGH
Total Fubing	~480,000-760,000	MODERATE
Peak (660s)	~600,000 soldiers	MODERATE

Geographic Distribution: One-third concentrated near Chang'an; two-thirds of empire had NO fubing units.

Jiedushi System (Post-749):

Period	Number of Jiedushi	Notes
742	10 frontier commands	9 headed by jiedushi
755	~10	An Lushan controlled 3
763	~40	Proliferated post-rebellion
820	~48	Many functionally independent

5.2 Major Battles

Battle of Talas (751 CE):

Force	Size	Casualties	Confidence
Tang	30,000 (10,000 Tang + 20,000 Karluk)	10,000-20,000 killed; 20,000 captured	MODERATE
Abbasid	30,000-40,000 (Western estimate)	~10,000 killed	MODERATE
Outcome	Decisive Abbasid victory; Karluk betrayal day 4-5	—	HIGH

Korean Campaigns:

Campaign	Tang Forces	Outcome
645 CE (1st invasion)	113,000 total	Failed; withdrew
667-668 CE	Major invasion	Conquered Goguryeo

5.3 Military Reforms Timeline

Date	Reform	Significance
677	Permanent frontier troops replace rotation	End of fubing rotation
711	First jiedushi established	Military governor system begins
737	Irregular troops replaced by permanent	Major professionalization
749	Fubing system abolished	Complete transition

6. Data Quality Assessment

High Confidence Data

- Official census figures (609, 742, 754)
- Tang Code specifications (502 articles, 653 CE)
- Kaiyuan Tongbao specifications
- Salt monopoly revenues post-758
- Emperor list and reign dates

Moderate Confidence Data

- Urban population estimates (Chang'an, Luoyang)
- Urbanization rates (Rozman 1973)
- Examination system statistics
- Battle force sizes (Chinese sources)

Low Confidence Data / Major Gaps

- All GDP estimates - methodology questioned
- Silk Road trade volumes - no quantitative data
- An Lushan death toll - 13M to 36M range

- Interest rate data - essentially absent
 - Regional economic disparities
 - Real wage data
 - Price indices
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Key Sources

- **Maddison, Angus** (2007). *Chinese Economic Performance in the Long Run*. OECD Development Centre
 - **von Glahn, Richard** (2016). *The Economic History of China*. Cambridge University Press
 - **Deng, Kent & O'Brien, Patrick** (2016). "China's GDP per capita from the Han Dynasty to Communist times." LSE Working Paper 229
 - **Twitchett, Denis** (ed.). *Cambridge History of China*, Vol. 3
 - **Lewis, Mark Edward**. *China's Cosmopolitan Empire: The Tang Dynasty*
 - **Johnson, Wallace** (1979). *The T'ang Code*. Princeton University Press
 - **Adshead, S.A.M.** *Tang China: The Rise of the East in World History*
 - **Rozman, Gilbert** (1973). *Urban Networks in Ch'ing China and Tokugawa Japan*
 - **White, Matthew** (2011). *Atrocity*
-

APPENDIX B: Abbasid Caliphate Quantitative Data (750-1258 CE)

Executive Summary

The Abbasid Caliphate represents the apex of medieval Islamic civilization, with GDP per capita reaching **\$890-990** (1990 Int\$) in 8th-century Iraq - approximately **3x subsistence level** and comparable to contemporary England. Baghdad grew to possibly **1-2 million inhabitants** (highly contested), while the caliphate's monetary system maintained remarkable stability until fragmentation. The 508-year dynasty saw **37 caliphs**, with effective power lost to the Buyids in **945 CE** and final destruction by Mongols in **1258 CE**. [Encyclopedia.com](https://www.encyclopedia.com)

1. Economic Indicators

1.1 GDP Estimates (Maddison 1990 International Dollars)

Period	Region	GDP per Capita	Confidence	Source
Rise (750-809)	Southern Iraq	\$890-990	MODERATE	Pamuk & Shatzmiller (2014)
Rise	Lower Egypt	\$800-910	MODERATE	Pamuk & Shatzmiller (2014)
Peak (809-861)	Iraq	~\$850-900	MODERATE	Estimate
Decline (post-1100)	Lower Egypt	\$590-680	MODERATE	Pamuk & Shatzmiller (2014)

Methodological Note: Estimates derive from "indirect method" linking real wages and urbanization to GDP per capita. Subsistence minimum: \$275-300. Peak Iraqi GDP was ~3x subsistence.

Comparative Context:

Polity	Period	GDP per capita
Abbasid Iraq (peak)	8th-9th c.	\$890-990
England	1250s	~\$800-850
Byzantine Empire	c.1000	~\$650-700
Roman Egypt	1st c. CE	~\$600-700

1.2 Trade Routes and Infrastructure

Indian Ocean Trade (Primary maritime network):

- Route established as regular by **mid-9th century**
- Journey time: ~4-6 months one way (monsoon-dependent)
- Main ports: **Basra, Ubullah, Siraf** (Persian Gulf) → India → Canton (China)
- Port of Siraf destroyed by earthquake 977 CE

Trade Goods:

Direction	Commodities
TO China	Ivory, pearls, dates, glass, textiles
FROM China	Silk, ceramics/porcelain, spices

DATA GAP: No comprehensive trade volume statistics survive. Archaeological evidence (shipwrecks) shows "large volume" but cannot be quantified.

1.3 Urbanization

Region	Period	Urban %	Confidence
Iraq	8th-9th c.	>20%	MODERATE
Egypt	8th-10th c.	~15-20%	LOW

1.4 Major Economic Centers

City	Peak Population	Period	Confidence
Baghdad	700,000-1,200,000	9th-10th c.	LOW-MODERATE
Basra	200,000-600,000	8th-9th c.	LOW
Samarra	100,000-200,000	836-892	LOW
Córdoba (rival)	250,000-500,000	10th c.	MODERATE

2. Demographic Indicators

2.1 Total Caliphate Population

Date	Population	Phase	Confidence	Notes
750 CE	45-55 million	Rise	LOW	Scholarly inference
~1000 CE	~60 million	Fragmentation	MODERATE	Encompasses Spain to Central Asia

DATA GAP: No direct population figures exist for 750 CE or 850 CE. Medieval Islamic sources focus on political events rather than demographics.

2.2 Baghdad Population (Critical Contested Data)

Period	Population	Confidence	Source
762 CE (founding)	Small settlement	HIGH	Historical consensus
~800 CE (Harun al-Rashid)	175,000-500,000	LOW-MODERATE	Conflicting estimates
~900 CE (peak)	1-2 million	LOW	Popular sources; disputed
1258 (pre-siege)	~1 million (claimed)	LOW	Contested

SCHOLARLY DISPUTE:

- **Higher estimates (1-2 million):** Popular histories, Wikipedia
- **Lower estimates (175,000-500,000):** Lapidus (300,000-500,000); Ian Morris (175,000 for 800 CE)
- **Archaeological critique** (Derek Kennet): Density calculations suggest high figures exaggerated; 350-900/ha densities required would be "extraordinarily high" vs. ~200/ha typical (Ucr)

Context: Baghdad's settled area was 550-860 hectares. At 200/ha density, maximum population would be ~110,000-172,000.

2.3 Mongol Conquest Mortality (1258 CE)

Source	Death Toll	Confidence
Hulegu's own estimate	~200,000	MODERATE (primary source)
Hamdallah Mustawfi (1330)	800,000	LOW
Martin Sicker	~90,000	MODERATE
Scholarly consensus	90,000-200,000	MODERATE

Context:

- Siege: January 29 - February 10, 1258
- Massacre continued ~1 week post-surrender

- Hulegu's army: ~138,000 Mongols ([Wikipedia](#))
- Baghdad defenders: ~30,000 troops
- Christians spared (Hulegu's wife Doquz Khatun was Christian)

2.4 Other Crisis Mortality

Event	Period	Impact	Confidence
Zanj Rebellion	869-883	500,000-2,500,000 (claims); likely tens to hundreds of thousands	LOW-MODERATE
Plague of Sheroe	627-628	Devastating to Sasanian Persia (pre-Abbasid)	MODERATE

3. Financial Indicators

3.1 Gold Dinar Standards

Characteristic	Value	Confidence
Weight (mithqal)	4.25 grams	HIGH
Purity (Umayyad reform, 696)	~97% fine gold	HIGH
Fatimid era purity	~99%+ fine gold	HIGH
Stability	Remarkably stable 750-950 CE	HIGH

3.2 Silver Dirham Standards

Characteristic	Value	Confidence
Nominal weight	2.97 grams (ideal)	HIGH
Actual average	2.71-2.8 grams	HIGH
Purity (early Abbasid)	~94-97% fine silver	MODERATE
Weight ratio to dinar	7 mithqals = 10 dirhams	HIGH

3.3 Exchange Rates

Period	Dinar:Dirham Ratio	Confidence
Early Abbasid (750-850)	1:20 (initial)	MODERATE
Mid-Abbasid	~1:13-20 (variable)	MODERATE
Later periods	"Fluctuated violently" (Ashtor)	HIGH

3.4 Currency Debasement Patterns

Period	Pattern	Confidence
717-721 CE	Abrupt <i>increase</i> in silver fineness at Wasit mint	HIGH
Early Abbasid (750-850)	Generally maintained high standards	MODERATE
Post-945 (fragmentation)	Regional mints with varying standards	MODERATE
Black dirhams (Bukharkhuda)	Severe Central Asian debasement	MODERATE

Key finding: Mints often **decreased or suspended output** rather than debasing during silver shortages.

3.5 Fiscal Data - Harun al-Rashid Era (785 CE)

Cash Collections:

- **313,780,000 dirhams** (~\$93.5 million silver grams)
- **3,816,000 dinars** (~16.2 million gold grams)

In-Kind Collections (same year):

- 4,000 pack animals from Khurasan carrying 1,000 silver ingots
- 1,000 slaves; 27,000 garments
- 30,000 myrobalan, 30,000 pounds sugar (from Ahwaz)
- Large amounts of raisins, salted fish, cumin, honey

Regional Revenue:

Province	Revenue	Period	Confidence
Egypt (pre-autonomous)	18 million dinars/year	Early Abbasid (CMU)	LOW-MODERATE
Tulunid Egypt tribute	300,000 dinars/year	868-905	MODERATE
Basra port taxes	20,000 dinars/year	Peak (Turcomat)	LOW

3.6 Tax Rates

Jizya (Poll Tax on Non-Muslims):

Wealth Class	Annual Rate
Poor	1 dinar (12 dirhams)
Middle	2 dinars (24 dirhams)
Rich	4 dinars (48 dirhams)

Note: 1 dinar ≈ 1 month's family subsistence; could buy ~1 sheep or 250-350 kg wheat

3.7 Interest Rates (Despite riba prohibition)

Period	Islamic Middle East	Europe
Until 10th century	6⅔% - 10%	33.3% (Uwo)
12th century (Saladin's Egypt)	10% - 15%	Lower

Source: Ashtor, Geniza documents

4. Institutional Indicators

4.1 Complete Caliph List (750-1258)

#	Caliph	Reign	Years	Succession	Phase
1	al-Saffah	750-754	4	Founder	Rise

#	Caliph	Reign	Years	Succession	Phase
2	al-Mansur	754-775	21	Peaceful	Rise
3	al-Mahdi	775-785	10	Peaceful	Rise
4	al-Hadi	785-786	1	Peaceful	Rise
5	Harun al-Rashid	786-809	23	Peaceful	Rise
6	al-Amin	809-813	4	Peaceful	Peak
7	al-Ma'mun	813-833	20	VIOLENT (civil war)	Peak
8	al-Mu'tasim	833-842	9	Peaceful	Peak
9	al-Wathiq	842-847	5	Peaceful	Peak
10	al-Mutawakkil	847-861	14	Peaceful	Peak
11-37	[26 more caliphs]	861-1258	—	Mixed	Fragmentation

Total: 37 caliphs over 508 years
Confidence: HIGH

4.2 Succession Statistics by Phase

Phase	Period	Caliphs	Avg. Reign	Violent %
Rise	750-809	5	11.8 years	0%
Peak	809-861	6	8.7 years	17%
Fragmentation	861-1258	26	15.3 years	46%
Overall	750-1258	37	13.7 years	35%

4.3 Major Succession Crises

Fourth Fitna (809-813):

- Cause: Harun's unusual succession plan splitting empire (802)
- Duration: 4 years
- Outcome: Siege of Baghdad (812-813); al-Amin executed

- First Abbasid caliph killed by another Abbasid
- **Confidence:** HIGH (documented in al-Tabari)

Anarchy at Samarra (861-870):

- 4 caliphs murdered in 9 years
- al-Mutawakkil assassinated by Turkish guards (861)
- al-Musta'in deposed and killed (866)
- al-Mu'tazz deposed and killed (869)
- al-Muhtadi assassinated (870)
- **Confidence:** HIGH

4.4 Timeline of Caliphal Power Loss

Date	Event	Significance
833	Turkish ghilman introduced	Military dependence begins
836	Capital moved to Samarra	Isolation from Baghdad
861-870	Anarchy at Samarra	Caliphate nearly collapses
934-936	Amir al-umara position created	Executive power ceded
945	Buyid takeover	Caliphs become figureheads
1055	Seljuk takeover	Temporal power to sultans
1136-1160	al-Muqtafi's revival	Partial restoration
1258	Mongol sack	End of caliphate

4.5 Four Sunni Madhhabs

School	Founder	Dates	Founded In	Confidence
Hanafi	Abu Hanifa	699-767 CE	Kufa, Iraq	HIGH
Maliki	Malik ibn Anas	711-795 CE	Medina	HIGH
Shafi'i	al-Shafi'i	767-820 CE	Egypt	HIGH

School	Founder	Dates	Founded In	Confidence
Hanbali	Ahmad ibn Hanbal	780-855 CE	Baghdad	HIGH

All four formalized by ~850 CE (250 AH).

4.6 Major Hadith Collections (Abbasid Era)

Collection	Compiler	Death Date	Hadith Count
Sahih al-Bukhari	al-Bukhari	870 CE	7,563 (2,600 unique)
Sahih Muslim	Muslim ibn al-Hajjaj	875 CE	~12,000 (4,000 unique)
Sunan Abu Dawud	Abu Dawud	889 CE	5,274
Sunan al-Tirmidhi	al-Tirmidhi	892 CE	~4,400
Sunan al-Nasa'i	al-Nasa'i	915 CE	5,758
Sunan Ibn Majah	Ibn Majah	887 CE	4,341

All compiled 840-912 CE during Abbasid era. **Confidence:** HIGH

4.7 Mihna (Inquisition) 833-848 CE

Parameter	Data	Confidence
Duration	15-18 years	HIGH
Initiated by	al-Ma'mun (833)	HIGH
Ended by	al-Mutawakkil (848)	HIGH
Doctrine tested	Createdness of Qur'an	HIGH
Most famous victim	Ahmad ibn Hanbal	HIGH

Significance: Failure marked end of caliphal role in defining religious orthodoxy; elevated traditionalist scholars.

4.8 Administrative Structure

Diwan System:

Department	Function
Diwan al-Kharaj	Taxation
Diwan al-Jund	Military affairs
Diwan al-Barid	Postal/intelligence
Diwan al-Rasa'il	Chancery
Diwan al-Mazalim	Grievances

Barid (Postal System): 930 relay stations; spacing 2-4 farsakhs (6-12 miles); speed up to ~100 miles/day

5. Military Indicators

5.1 Army Sizes by Phase

Rise Phase (750-809):

Force	Size	Confidence
Total callable force	125,000	MODERATE
Early professional army	"Tens of thousands"	LOW

Peak Phase - Samarra Period (809-861):

Force	Size	Source	Confidence
Total Samarra army	94,353	Derek Kennet (archaeological)	MODERATE
Alternative estimate	20,000-30,000	Helmut Töllner	MODERATE
Turkish slave soldiers	10,000-15,000 (~50%)	Töllner	MODERATE
Khurasani cavalry/infantry	4,000-5,000	Hugh Kennedy	MODERATE

Fourth Fitna Civil War (811-813):

Force	Size	Casualties	Confidence
al-Amin's army (Rayy, 811)	40,000-60,000	Heavy	HIGH
al-Ma'mun's force (Tahir)	3,800-5,000	Minimal	HIGH
Total Fitna casualties	—	~70,000 killed	LOW

Fragmentation (861-1258):

Period	Force	Size
1258	Baghdad garrison	~30,000
1258	Mongol army	~138,000 (largest ever)

5.2 Battle of Talas (751 CE)

Force	Size	Casualties	Confidence
Abbasid	30,000-40,000	~10,000 killed	MODERATE
Tang	10,000 Tang + 20,000 Karluk	20,000-30,000 killed	MODERATE
Outcome	Decisive Abbasid victory	—	HIGH

Duration: 4-5 days. Karluk defection decisive.

5.3 Zanj Rebellion (869-883 CE)

Metric	Estimate	Confidence
Duration	14-15 years	HIGH
Rebel force (claimed)	Up to 500,000	LOW
Death toll (sources)	500,000-2,500,000	LOW
Death toll (realistic)	Tens to hundreds of thousands	MODERATE

Impact: Devastating to southern Iraq; Basra never fully recovered.

5.4 Byzantine Frontier Garrisons

Fortress	Garrison	Period	Confidence
Mopsuestia	4,000	Early Abbasid	MODERATE
Hadath	4,000	Early Abbasid	MODERATE
Malatya	4,000	757/758 CE	MODERATE

786 CE: Harun al-Rashid established second defensive line (al-'Awasim)

5.5 Turkification Timeline

Date	Development
750-833	Khurasani dominance
820s-830s	Turkish ghulam introduction (al-Ma'mun/al-Mu'tasim)
836	Capital moved to Samarra (to separate Turks from Baghdad)
833-870	Turkish dominance; Turks murdered al-Mutawakkil (861)

6. Data Quality Assessment

High Confidence Data

- Caliph list and reign dates
- Dinar/dirham weight standards
- Madhhab formation dates
- Hadith collection dates and sizes
- Mihna dates
- Battle of Talas outcome
- Mongol conquest date

Moderate Confidence Data

- GDP per capita estimates (Pamuk & Shatzmiller methodology)
- Fiscal revenue figures (785 CE)
- Army sizes (Samarra archaeological survey)
- Byzantine frontier garrisons
- Fourth Fitna army sizes

Low Confidence Data / Major Gaps

- **Baghdad population** - 175,000 to 2 million range
- **Total caliphate population** - single estimate (~60M for 1000 CE)
- **Trade volumes** - qualitative only
- **Naval forces** - almost no data
- **Defense spending %** - not available
- **Provincial armies** - limited data
- **Agricultural yields** - no statistics survive
- **Tax as % of GDP** - cannot calculate

Comparative Analysis: Tang vs. Abbasid

Key Metrics Comparison Table

Indicator	Tang Dynasty	Abbasid Caliphate	Confidence
Duration	289 years (618-907)	508 years (750-1258)	HIGH
GDP per capita	~\$480 (Maddison estimate)	\$890-990 (peak Iraq)	LOW-MODERATE
Peak population	50-75 million	~60 million (c.1000)	MODERATE
Largest city	Chang'an (1-2 million)	Baghdad (1-2 million)	LOW
Total rulers	20-21 emperors	37 caliphs	HIGH
Average reign	12.6 years	13.7 years	HIGH

Indicator	Tang Dynasty	Abbasid Caliphate	Confidence
Violent successions	~40%	35%	MODERATE
Legal codification	Tang Code (502 articles, 653 CE)	4 madhhabs + 6 hadith collections	HIGH
Currency	Copper cash (3.21-3.75g)	Gold dinar (4.25g) / Silver dirham (2.97g)	HIGH
Critical turning point	An Lushan Rebellion (755)	Buyid takeover (945)	HIGH
Battle between empires	Talas 751: Tang lost	Talas 751: Abbasid won	HIGH
Final collapse cause	Internal fragmentation	Mongol conquest (1258)	HIGH

Institutional Sophistication

Feature	Tang	Abbasid
Civil service exam	Yes (keju)	No formal exam
Codified law	Yes (Tang Code)	Religious law (Sharia)
Centralized bureaucracy	Strong (3 Depts, 6 Ministries)	Strong (Diwan system)
Provincial control	Declining post-755	Declining post-861
Military professionalization	749 (fubing → jiedushi)	833 (ghulam system)

Economic Structure

Feature	Tang	Abbasid
Primary revenue source	Salt monopoly (>50% by 780)	Land tax (kharaj) + poll tax (jizya)
Trade orientation	Silk Road (overland)	Indian Ocean (maritime) + Silk Road
Currency stability	Declining post-755	Stable until ~950
GDP methodology	Backward extrapolation only	Wage-based indirect method

Methodological Notes

Sources Hierarchy

1. **Primary sources** (al-Tabari, Tang census records, numismatic evidence): Highest reliability for specific dates and events
2. **Modern economic history** (Pamuk & Shatzmiller, Maddison): Methodologically rigorous but dependent on assumptions
3. **Traditional histories** (medieval chronicles): Valuable for events, unreliable for numbers
4. **Popular syntheses**: Useful for context, verify all figures

Key Scholarly Disputes

1. **Maddison GDP estimates**: Challenged by Deng & O'Brien (2016) as "not fit for purpose"
2. **Baghdad population**: Archaeological vs. literary evidence conflict
3. **An Lushan death toll**: Census loss (36M) vs. actual deaths (13M)
4. **Watson's "Arab Agricultural Revolution"**: Contested by Decker (2009)

Data Gap Patterns

Both empires suffer from:

- No direct GDP measurements possible
 - Population figures from administrative records, not demographic censuses
 - Trade volumes qualitative only
 - Military casualty figures routinely exaggerated
 - Late-period data worse than early-period
-

Key Sources

Tang Dynasty

- **Twitchett, Denis** (ed.). *Cambridge History of China*, Vol. 3
- **Lewis, Mark Edward**. *China's Cosmopolitan Empire: The Tang Dynasty*
- **Johnson, Wallace** (1979). *The T'ang Code*. Princeton UP

- **Adshead, S.A.M.** *Tang China: The Rise of the East in World History*
- **Maddison, Angus** (2007). *Chinese Economic Performance in the Long Run*. OECD
- **Graff, David.** *Medieval Chinese Warfare*
- **von Glahn, Richard** (2016). *The Economic History of China*. Cambridge UP

Abbasid Caliphate

- **Kennedy, Hugh** (2001). *The Armies of the Caliphs*
- **Kennedy, Hugh** (2005). *When Baghdad Ruled the Muslim World*
- **Pamuk, Şevket & Shatzmiller, Maya** (2014). "Plagues, Wages, and Economic Change." *Journal of Economic History*
- **Shatzmiller, Maya.** Medieval Islamic Economy Database (UWO)
- **Hodgson, Marshall.** *The Venture of Islam*
- **Abu Yusuf** (786 CE). *Kitab al-Kharaj*
- **Ashtor, Eliyahu** (1969/1976). *Prix et Salaires; Social and Economic History*
- **Wickham, Chris** (2005). *Framing the Early Middle Ages*

Compiled December 2025 for Infonomics Project CAS Framework. All confidence levels reflect current scholarly consensus and source limitations. Medieval quantitative data should be treated as approximations subject to revision as new scholarship emerges.